Kartick Samanta

Integrated Ph.D. Scholar in Applied Mathematics, TIFR Centre for Applicable Mathematics

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RESEARCH INTERESTS

Geophysical Fluid Dynamics, Numerical Analysis and Scientific Computing, Machine Learning, Deep Learning.

EDUCATION

Integrated Ph.D. (M.Sc. + Ph.D.) in Applied Mathematics — Jan 2024 – Present
TIFR Centre for Applicable Mathematics (TIFR-CAM) — *Bengaluru, India*

- Currently in the third year.
- Research guide: Dr. Jim Thomas.

B.Sc. in Mathematics (CGPA 9.39/10) — Aug 2021 – Jun 2024
Ramakrishna Mission Vivekananda Centenary College — *Rahara, Kolkata, India*

Higher Secondary (Science) (93.8%) — Jun 2019 – Jun 2021
Boruna Satsanga High School — *Ghatal, West Bengal, India*

Secondary (91.28%) — Jan 2013 – Mar 2019
Nabin Manua Iswar Chandra High School — *Ghatal, West Bengal, India*

RESEARCH EXPERIENCE

Neural Networks for Nonlinear Dispersive Wave Turbulence — Ongoing
Research project, TIFR-CAM (guide: Dr. Jim Thomas) — *Bengaluru*

PROJECTS

- **Spectral and Numerical Solvers for PDEs** — finite-difference and spectral discretizations of one-dimensional transport and diffusion equations: explicit and implicit time stepping, Runge–Kutta methods, Fourier methods and Chebyshev differentiation.
- **Deep Learning with PyTorch** [GitHub] — ANN and CNN classifiers on Fashion-MNIST, U-Net, autoencoder, GAN and Fourier neural operator; custom dataset and dataloader classes and a reusable training pipeline with validation and checkpointing; function-approximation experiments on depth, width and extrapolation.
- **Neural Network from Scratch** [GitHub] — multilayer network in Python and NumPy with no deep learning framework: forward propagation, backpropagation and gradient-descent training.
- **Numerical Methods and Numerical Linear Algebra** — Python implementations of classical algorithms: root finding (bisection, Newton–Raphson, secant), numerical integration and differentiation, interpolation, curve fitting and numerical linear algebra.
- **PDF Bookmark Manager** [GitHub] — desktop app in Python with PySide6 and PyMuPDF to preview PDFs, add and edit chapter and section bookmarks, and export bookmarked PDFs; released as a Windows executable built with PyInstaller.

Earlier projects:

- Face recognition using eigenfaces (PCA) in MATLAB.
- Fourier transform in audio compression.
- Linear transformations in computer graphics.
- Random number generation and statistical tests of fairness.
- Burnside's theorem, bridging combinatorics and group theory.

TECHNICAL SKILLS

Programming	Python, C, C++, R
Math software	MATLAB, Mathematica, SageMath, GeoGebra
Scientific Python	NumPy, SciPy, SymPy, pandas, Matplotlib, h5py, CuPy
Machine learning	PyTorch; Fourier neural operators, diffusion models (DDPM, PDE-Refiner), U-Net, GAN, autoencoders
Numerical methods	Fourier and Chebyshev spectral methods, dealiasing, finite differences, explicit, implicit and adaptive Runge–Kutta time stepping, operator splitting, convergence testing, numerical linear algebra
HPC	Linux clusters over SSH, Slurm , OpenMP ; parallel file systems, tmux
GPU computing	CUDA, GPU-accelerated simulations
Development tools	VS Code, Jupyter Notebook and JupyterLab, Google Colab, Anaconda, Git and GitHub
Operating systems	Linux (Ubuntu), macOS, Windows; Linux terminal, PowerShell, Command Prompt
Software and web	Flask, PySide6/PyQt, Tkinter, PyMuPDF, PyInstaller; HTML, CSS, MySQL
Documents	LaTeX (Overleaf, MiKTeX with TeXstudio), Markdown; Microsoft Word, Excel, PowerPoint, Access
Design and media	Adobe Photoshop, Premiere Pro, Audition; Audacity; Canva
Soft skills	Analytical and critical thinking, problem solving, teaching

ADVANCED COURSEWORK AND INDEPENDENT STUDY

- **IISc Bengaluru (audit, 2026)** — *Introduction to Data Science*; *Introduction to Scalable Systems*.
- **Geophysical Fluid Dynamics(TIFR-CAM)** — Reading Course.

SCHOOLS, WORKSHOPS AND CERTIFICATIONS

- Mathematics Training and Talent Search (MTTS) Summer Camp, IIT Dhanbad, 2023.
- MTTS Init-Math, Ramakrishna Mission Vivekananda Centenary College, Rahara, 2023.
- Python certification, Spoken Tutorial Project, IIT Bombay.

LANGUAGES

Bengali (native) | English (full professional) | Hindi (full professional)